

1 SCM Case Summary: idealized_sbl

This section was generated automatically from the SCM diagnostics artifact `results/idealized_sbl_smoke/summary.json`. The simulation completed with `OrdinaryDiffEqRosenbrock.Rodas5P{0, ADTypes.AutoFiniteDiff{Val{:forward}, Val{:forward}, Val{:hcentral}}, Nothing, Nothing, Int64}, Nothing, typeof(OrdinaryDiffEqCore.Val{:forward}()), true, nothing, typeof(OrdinaryDiffEqCore.trivial_limiter!), typeof(OrdinaryDiffEqCore.trivial_limiter!)}` and returned `Success`.

1.1 Core Run Metadata

- Case identifier: `idealized_sbl`
- Output directory: `results/idealized_sbl_smoke`
- Number of saved time samples: 13
- Number of profile snapshots: 1
- Solver accepted/rejected steps: 65 / 2
- RHS evaluations: 16333
- Solver tolerances: $\text{abstol} = 1.000 \times 10^{-8}$, $\text{reitol} = 1.000 \times 10^{-6}$

1.2 Numerical Verification

- Maximum surface energy closure error: 0 W m^{-2}
- Fold-proximity fraction: 0.000000%
- Diffusivity bounds: $K_m \in [0.184562, 108.426452] \text{ m}^2 \text{ s}^{-1}$
- Richardson number bounds: $Ri \in [0.034185, 34.722193]$

1.3 Generated Artifacts

- Time-series CSV: `results/idealized_sbl_smoke/time_series.csv`
- Diagnostic summary JSON: `results/idealized_sbl_smoke/summary.json`
- Payload JLD2: `not generated`

1.4 Figure Manifest

1. Time series of T_s , sensible heat flux, and friction velocity
2. Time-height contour of wind speed with LLJ evolution
3. Time-height contour of potential temperature and inversion growth
4. Vertical profiles of U , θ , and K_m at representative times
5. Surface energy budget components (R_n , H , G , storage)
6. Phase portrait on regularized manifold (Δ vs e_{xi})
7. Eddy diffusivity response (K_m , K_h) versus stability/Richardson number
8. Fold proximity diagnostics versus time

1.5 Included Plots